# MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIRST SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

**MG1DSCCHE100– Fundamentals of Chemistry-1**

Duration: 1.5 hrs. Maximum Marks: 50

*Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.*

**Part A**

Multiple Choice Questions

Answer **All** Questions

Each question carries **1** mark

1. Which of the following represents a spectral series in the hydrogen atom's emission line spectrum?

   a) Lyman series b) Rayleigh series c) Pauling Scale d) Pauli's series [K] [CO1]

2. What term describes a region around an atom's nucleus where there is a high probability of finding an electron?

   a) Energy level b) Orbital c) Proton cloud d) Electron path [K] [CO1]

3. Which among the following pair has a diagonal relationship

   a) Silicon and Boron b) Boron and Aluminium

   c) Lithium and Beryllium d) Sodium and Potassium [K] [CO5]

4. Choose the correct decreasing order of stability of carbanions

   a) primary > secondary > tertiary

   b) secondary > primary > tertiary

   c) tertiary > primary > secondary

   d) tertiary > secondary> primary [A] [CO3,4]

5. Which statement is true about hyperconjugation?

   a) Hyperconjugation involves the overlap of p-orbitals between adjacent atoms

   b) Hyperconjugation stabilizes carbocations by donating electron density from adjacent σ-bonds

   c) Hyperconjugation only occurs in aromatic compounds

   d) Hyperconjugation is a strong intermolecular force affecting boiling points [K] [CO3,4]

6. What is the maximum number of electrons allowed in an orbital

   2 b)3 c)1 d)4 [K] [CO1]

7. What is catenation

   a) cation formation b) anion formation

   b) c) bonding of atoms of same element into a chain d) chain shortening reaction [K] [CO2]

8. What is an example for metalloid?

   a) Carbon b) Sodium c) Boron d) Oxygen [K] [CO5]

9. What is an example of compound with covalent bond
   a) HCl b) NaCl c) KCl d) KBr [K] [CO5]

10. Which among the following has the highest electronegativity in Pauling's scale?
    a) Cl b) F c) Cs d) Ne [K] [CO5]

[1 x 10 = 10]

**Part B**

Short Answer Type Questions

Answer **4** Questions. Each question carries **3** marks

11. Explain Heisenberg's uncertainty principle [U] [CO1]
12. Why do Cu and Cr deviate from the expected electronic configurations of their respective groups? [A] [CO1]
13. Illustrate homolysis and heterolysis in C-Cl bond in $CH_3$-Cl, using curved arrows [A] CO[3,4]
14. Draw and explain the structure of carbocation [U] [CO3,4]
15. Differentiate between valency and oxidation state [U] [CO5]
16. What are the reasons behind the decrease in atomic radii across a period in the periodic table? [A] [CO5]

[3 x 4 = 12]

Part **C**

Short Essay Type Questions

Answer **4** Questions. Each question carries **7** marks

17. How does the inductive effect influence the acidity of carboxylic acids, and what role do electron-withdrawing and electron-donating groups play in this process? [A] [CO3,4]
18. Calculate the effective nuclear charge ($Z_{eff}$) for the outermost electrons in Li, B, N and Ne. [A] [CO5]
19. What are quantum numbers? Discuss the significance of quantum numbers. [U] [CO1]
20. What are the features of hybridisation? Explain hybridisation of Carbons in ethane and ethene [U] [CO2]
21. Explain the atomic spectra of Hydrogen atom using Bohr theory [U] [CO1]
22. Define ionisation enthalpy? Explain how it varies along a period and in a group? [U] [CO5]

[7 x 4 = 28]